

Projected-Krylov reconstruction of frustrated signs with fixed-node amplitude feedback

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We study the sign structure of the spin-1/2 square-lattice J_1 - J_2 Heisenberg model near maximal frustration, focusing on $J_2/J_1 = 1/2$. Writing the wave function as $\psi(x) = s(x)a(x)$ with $a(x) > 0$, we separate the reconstruction problem into a sign update and a positive-amplitude update. A single projected Krylov step provides an accurate, label-free reconstruction of the physically important sign corrections when supplied with a sufficiently accurate amplitude, while a lattice fixed-node projector supplies sign-free information about the positive amplitude for fixed signs.

On exact 4×4 systems the coupled fixed-node/Krylov map can recover the exact ground-state signs without target-ground-state information entering the updates. On 6×6 and 8×8 , one-step Krylov signs substantially improve fixed-node energies over Marshall signs and remain competitive with high-accuracy variational benchmarks. The principal scaling bottleneck is not the sign rule itself but the handoff from fixed-node walkers to a callable positive amplitude. We show that independent Euclidean maximum-likelihood gradients from fixed-node replicas can be strongly anti-aligned, whereas a matrix-free stochastic-reconfiguration update rotates the direction into one that improves an independent held-out replica. Repeated learned updates remain non-monotone in the physical fixed-node objective, motivating a physically gated outer feedback loop.

I. INTRODUCTION

Frustration in the square-lattice J_1 - J_2 Heisenberg antiferromagnet produces a nontrivial many-body sign structure precisely where unbiased Monte Carlo methods are most difficult to apply. Instead of attempting to remove the sign structure globally, we ask whether its two components can be treated by different numerical mechanisms:

$$\psi(x) = s(x)a(x), \quad s(x) = \pm 1, \quad a(x) > 0. \quad (1)$$

The central separation in this work is that the sign s is updated by a projected Krylov rule, whereas positive-amplitude information is supplied by a sign-free fixed-node projection.

This leads to the feedback architecture

$$s_k \xrightarrow{\text{FN}} a_{k+1} \xrightarrow{\text{K1}} s_{k+1}, \quad (2)$$

where the first arrow denotes a positive fixed-node solve at fixed signs and the second denotes a sign-changing projected-Krylov update. The two arrows must not be conflated: Krylov reconstruction supplies signs, not the new positive amplitude.

The paper is organized around three questions. First, how much of the frustrated sign structure is captured by one Krylov step? Second, does the coupled fixed-node/Krylov map self-correct on controlled systems? Third, what prevents the same feedback from scaling to larger lattices?

II. PROJECTED-KRYLOV RECONSTRUCTION OF THE SIGN STRUCTURE

For a positive amplitude $a(x)$ and a reference sign pattern $s(x)$, define the local Krylov field

$$r_s(x) = \frac{H[as](x)}{a(x)s(x)}. \quad (3)$$

A one-step Krylov vector obeys

$$[(T - H)as](x) = [T - r_s(x)]a(x)s(x), \quad (4)$$

so retaining only its sign gives

$$s'(x) = s(x) \text{sgn}[T - r_s(x)]. \quad (5)$$

The threshold T is inferred without target sign labels, using either fixed-amplitude energy minimization on exact small systems or robust two-cluster thresholding on sampled larger systems.

Figure 1 shows the exact 4×4 result. At $J_2/J_1 = 0.5$, Marshall signs have $O_S = 0.974538$, while one K1 update reaches $O_S = 0.999319$. The same step reduces the fixed-amplitude energy error per site from 1.16×10^{-2} to 5.51×10^{-4} . The one-step reconstruction remains effective over a broad frustration range provided the reference sign convention lies in the appropriate sign basin.

On 6×6 , an independent fixed-amplitude comparison using the same ViT modulus gives

$$E_{K1} - E_{\text{ViT}} = +0.00420(1035), \quad (6)$$

$$E_{K1} - E_M = -0.17247(4489), \quad (7)$$

in total energy. Thus K1 does not resolve below the ViT sign energy at this precision, but it reproduces essentially

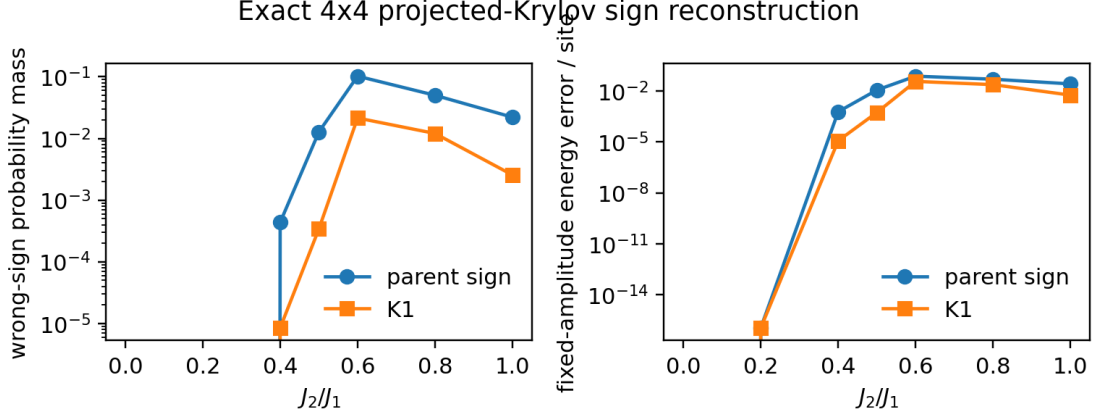


FIG. 1. Exact 4×4 benchmark of one-step projected-Krylov sign reconstruction. At each coupling the parent sign convention shown is the appropriate Marshall or stripe reference. The left panel shows the ground-state probability mass carrying the wrong sign; the right panel shows the corresponding fixed-amplitude variational energy error per site. Source: `square_exact_energyopt.csv`.

the same physical energy while substantially improving over Marshall signs.

The useful signal is primarily off diagonal. In the Marshall gauge the nearest-neighbor J_1 exchange contributes with the unfrustrated sign, while the frustrating J_2 exchange contributes oppositely. Their amplitude-weighted competition produces most of the observed K1 separation. On the 6×6 diagnostic sample, one K1 step removes approximately 95% of the physically weighted wrong-sign mass.

A naive second Krylov step does not continue this improvement. With the first-step amplitude distortion left in place, unsupervised K2 thresholds change only ~ 1 –2% of physical configurations yet raise the fixed-amplitude energy by 0.65–0.96 in total energy. We therefore use K1 as the sign reconstruction primitive and obtain subsequent amplitude information from fixed node rather than from an undamped Krylov continuation.

III. FIXED-NODE AMPLITUDE FEEDBACK

For fixed signs, the lattice fixed-node Hamiltonian is sign free and its projected ground state provides a positive amplitude $\phi_{\text{FN}}(x)$. The desired feedback is therefore Eq. (2), with each K1 field recomputed from the current amplitude and current sign state.

The exact 4×4 benchmark in Fig. 3 starts from the $J_2 = 0$ modulus and Marshall signs. No target- J_2 ground-state information enters either the fixed-node solve or the threshold update. The loop reaches the exact target sign pattern at iteration 100, with final amplitude fidelity $F_a = 0.9999946$ and guide-energy error 2.60×10^{-5} .

This exact result also clarifies an earlier apparent recurrence. A Marshall-referenced update repeatedly reconstructs a correction in a fixed coordinate and can stall

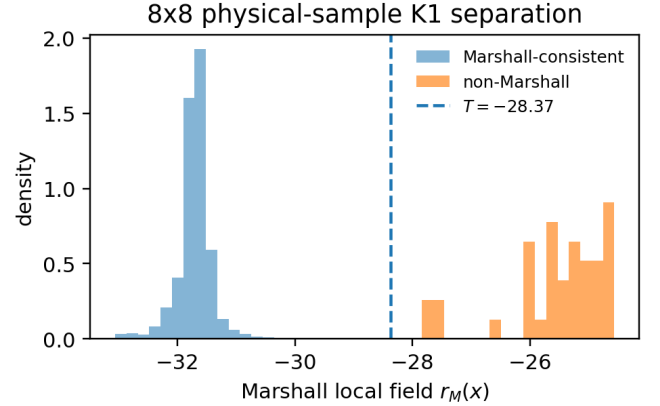


FIG. 2. Distribution of the Marshall local Krylov field on an 8×8 physical sample, separated using the stored ViT sign only as an offline diagnostic. The production threshold $T = -28.3711$ is inferred label-free. The two populations explain why a single scalar K1 step removes most of the physically relevant sign error.

near $O_S \simeq 0.9994$. The genuine nonlinear map must instead recompute r_{s_k} from the current sign state. In that formulation long plateaus are allowed: fixed node can improve the amplitude for many iterations before the next sign chamber becomes energetically accessible.

A. Mixed walkers and the ideal half step

Importance-sampled fixed-node walkers obey

$$f(x) \propto a_g(x)\phi_{\text{FN}}(x). \quad (8)$$

Exact 4x4 self-correcting current-sign loop

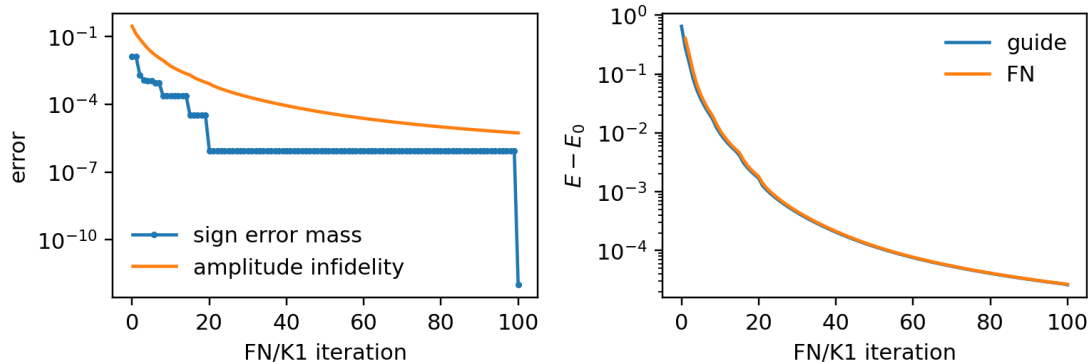


FIG. 3. Exact 4×4 current-sign fixed-node/K1 loop at $J_2/J_1 = 0.5$, initialized with Marshall signs and the exact $J_2 = 0$ ground-state modulus. Target- J_2 exact information is used only for the plotted diagnostics. Both the sign error and amplitude infidelity contract, with long plateaus followed by discrete sign-chamber changes.

If a Born model is trained by exact maximum likelihood, $p_\theta(x) \propto a_\theta(x)^2$, matching $p_\theta = f$ gives

$$a_{\text{new}}(x) \propto \sqrt{a_g(x)\phi_{\text{FN}}(x)}. \quad (9)$$

Exact 4×4 tests verify this geometric-mean half step. Starting from the exact target modulus with Marshall signs, the first half step followed by K1 reaches $E = -8.4457866$ from an initial guide energy -8.2729091 and has sign overlap 0.9994168 . A deliberately poor uniform-amplitude start is non-monotone at the first update but subsequently recovers, showing that Eq. (9) is a local amplitude mechanism rather than a global monotonicity theorem.

IV. SCALING AND THE AMPLITUDE-HANDOFF BOTTLENECK

A raw walker histogram is not a scalable callable amplitude. On exact 4×4 , millions of ideal independent samples are already required to approach the exact-amplitude K1 ceiling when each configuration amplitude is estimated independently. A four-bin density-ratio correction provides a controlled low-dimensional prototype and stabilizes the 4×4 loop with $\mathcal{O}(10^4)$ samples.

At 6×6 , increasing the fixed-node population removes the dominant replica disagreement and the tested four-bin loop reaches a practical fixed point: after the first amplitude handoff the K1 sign guide is unchanged on the physical diagnostic measure and the next fitted correction is an order of magnitude smaller than its pairwise noise. The same static construction is not computationally viable at 8×8 , where recomputing nested local energies generates tens of millions of cached states and hundreds of millions of ViT evaluations. This identifies the principal scalable object: a flat callable model for the positive amplitude.

The 8×8 K1-FN mean is $E/N = -0.498210$ from two $M = 128$ replicas, compared with -0.494961 for Marshall-FN. On the identical 8×8 periodic geometry, this places the K1-guided fixed-node result below the reported DMRG $D = 32000$ and FAMPS $D = 12000$ energies, while remaining above the RBM+PP and FAMPS $D = 24000$ benchmarks in Ref. [1]. Because the fixed-node estimate is itself population sampled, this comparison is used as a scale benchmark rather than a claim of precision ordering among methods.

V. SCALABLE AMPLITUDE LEARNING BY STOCHASTIC RECONFIGURATION

Direct warm-start maximum-likelihood training of the full ViT amplitude does not transfer between independent fixed-node replicas under ordinary Euclidean optimization. At the initial 8×8 guide, the two empirical MLE gradients have

$$\cos(g_1, g_2) = -0.8529. \quad (10)$$

This anti-alignment persists in an infinitesimal Euclidean line search and is therefore not merely an oversized Adam step.

Matrix-free stochastic reconfiguration changes the geometry. With empirical Fisher matrix S and diagonal shift $\lambda = 1$,

$$g_2^T (S + \lambda I)^{-1} g_1 = +3.85, \quad (11)$$

despite the negative Euclidean cosine. A finite line search then gives a held-out log-likelihood gain $+0.015575$ at $\eta = 0.01$, while retaining an importance-sampling ESS of $3236/4096$.

The corresponding amplitude refresh shifts the label-free K1 threshold from $T_0 = -28.37108$ to $T_1 = -28.82417$. Under the correct $a^{1.2}$ threshold measure,

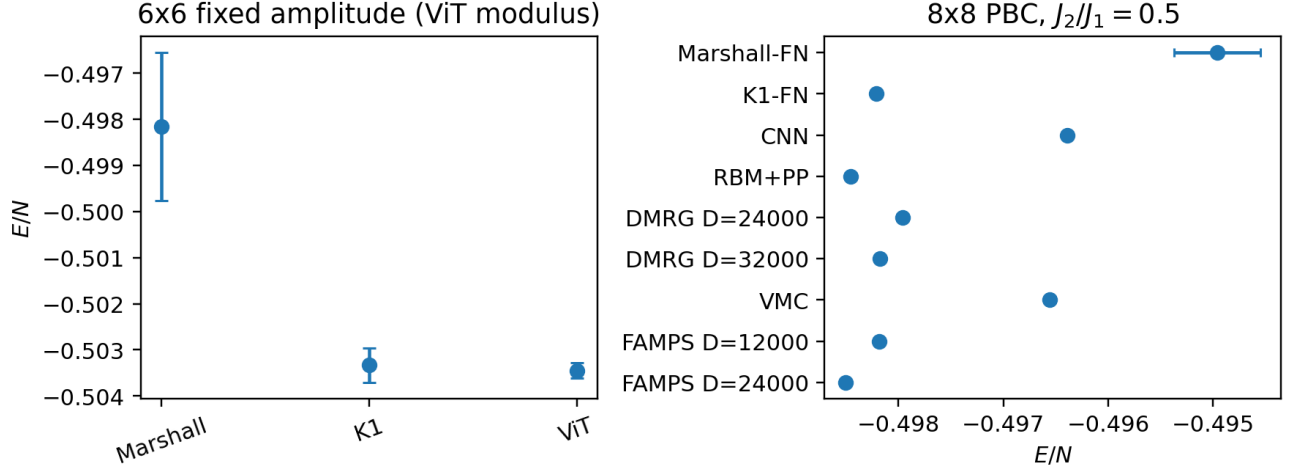


FIG. 4. Energy benchmarks. Left: 6×6 fixed-amplitude energies using the same ViT modulus and Marshall, K1, or ViT signs. Right: 8×8 periodic-boundary $J_2/J_1 = 0.5$ ground-state energy benchmarks. Literature values are from Qian and Qin, Ref. [1], Table I. The K1-FN and Marshall-FN error bars are two-replica spread proxies, not asymptotic statistical uncertainties.

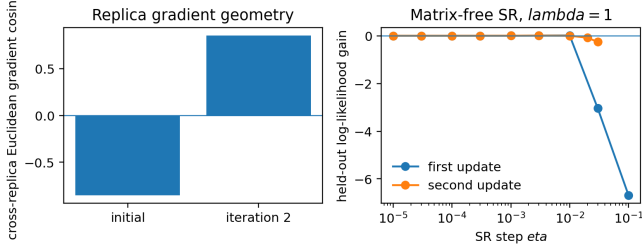


FIG. 5. Cross-replica amplitude-learning diagnostics on 8×8 . Left: Euclidean gradient cosine for the first and second learned updates. Right: held-out likelihood gain along the matrix-free SR direction at $\lambda = 1$. The validation-selected steps are $\eta = 0.01$ and 0.003 .

the K1 branch changes on 9.27% of the reweighted mass. Thus the learned amplitude update is not merely a likelihood perturbation; it generates a nontrivial sign refresh.

VI. CLOSED-LOOP CONVERGENCE AND PRESENT LIMITATION

A first pair of matched $M = 128$ fixed-node runs using the SR-updated amplitude and refreshed K1 signs gave total energies -31.89679 and -31.90567 . However, an independent reconstruction of a numerically indistinguishable SR checkpoint produced a second pair with mean -31.83896 , despite log-amplitude differences below 5×10^{-6} on a direct state audit. We therefore do not use the first pair alone as evidence for a statistically resolved energy improvement over the baseline K1-FN value -31.88542 .

The second learned SR step provides a complementary convergence diagnostic. Its independent-replica Eu-

clidean gradients become aligned, $\cos(g_1, g_2) = +0.8568$, and the validation-optimal SR step shrinks to $\eta = 0.003$. Nevertheless, after amplitude refresh and K1 reconstruction, the matched FN mean worsens to

$$E_{\text{iter}2} = -31.81537, \quad (12)$$

with a two-replica spread proxy 0.00947 . Improving the MLE objective is therefore not sufficient to accept an outer-loop update.

A frozen-sign diagnostic isolates this failure mode by holding the successful iteration-1 signs s_1 fixed while replacing a_1 by the iteration-2 amplitude a_2 . The first completed replica gives $E_{\text{FN}}[a_2, s_1] = -31.84646$, already worse than the corresponding $E_{\text{FN}}[a_1, s_1] = -31.89679$. The paired frozen-sign run is still in progress; the interpretation will be updated only after that control closes.

VII. DISCUSSION

The results support a clean division of labor. One projected Krylov step is a cheap extractor of the physically important frustrated sign structure. Fixed node supplies positive-amplitude information without reintroducing a sign problem. Their exact 4×4 composition is self-correcting, and the K1 sign guide remains effective through 8×8 .

The unresolved scaling problem is the amplitude hand-off. Static nonparametric or nested corrections become computationally expensive, while ordinary Euclidean neural updates are replica dependent. Matrix-free stochastic reconfiguration supplies the first transferable learned direction at 8×8 , but the physical projector must remain the acceptance criterion for repeated updates.

The natural next step is therefore not a more complicated sign model. It is a statistically controlled outer loop in which each learned amplitude step is validated on independent FN populations and accepted only if a matched fixed-node energy diagnostic is stable under replication. The current frozen-sign experiment further separates amplitude degradation from any failure of K1

itself.

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[TODO: Add funding, computing resources, and collaborators.]

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- [1] X. Qian and M. Qin, Augmenting density matrix renormalization group with disentanglers, Chinese Physics Letters **40**, 057102 (2023).